

"While there are challenges that have been clearly identified by industry stakeholders—especially related to the mode of selection of projects under the JNNM (the price bidding mechanism) and achieving project financial closure—the solar market will continue to grow both based on a real need for energy and driven by such key mechanisms as renewable purchase obligations," says Debasish Paul Choudhury, president, SEMI India.

"It is, however, important that the first phase of power plant projects under both the state and national policies get implemented in a timely manner, are well executed, and meet performance and quality norms," he adds.

Grid-connected farms: Main growth driver

"The solar PV market in India could be bigger than 1 GW in 2012. This growth will be driven primarily by grid-connected farms, with more than 90 per cent of the market being grid-connected. The country is rapidly developing indigenous capability to build a complete PV ecosystem. This would lead to faster scaling of the scope of the business," shares Vivek Chaturvedi, global head-sales and marketing, Moser Baer Solar.

In the last two decades, the solar energy industry in India has developed in off-grid mode since the policy framework to allow solar power to be connected to the grid did not exist until January 2008. The real fillip came in 2010 with the JNNM. As of March 31, 2011, 38 MW of grid-connected solar power plants have already been established, and this number is likely to rise dramatically by the end of 2011.

Manufacturing base: How big?

Solar cells are the basic elements of a solar module or panel. These are made of silicon—a common semiconductor—and typically measure 15×15 cm².

Globally, C-Si is the dominating technology with 80-85 per cent market share. We believe the same scenario will emerge in India as C-Si technology demonstrates a strong roadmap of cost and price decline in future years — Ravi Surapaneni, vice president, Solar Semiconductor



Currently, India has solar cell manufacturing capacity of 600-800 MW and module capacity of more than 1000 MW. If all the announcements made by new players are implemented, the cell capacity will double in the next twelve to 18 months. Indian companies are taking significant steps in developing systems business as well, on standalone and partnership basis with international players.

There are as many as 60 companies making solar modules, of which about 15 to 20 manufacture solar cells. Several manufacturers which began as module manufacturers have begun manufacturing solar cells themselves.

However, at present, solar PV cell manufacturing in India is totally dependent on imports for crucial materials like silicon wafers, encapsulating material and high transmission glass. Indigenous capabilities are needed on the material front in order to allow for manufacturing innovation and cost reduction.

Technology trends

Crystalline silicon (C-Si) solar cell technology continues to be the market leader, accounting for 85 per cent of the global market.

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demonstrates a strong roadmap of cost and price decline in future years," says Ravi Surapaneni, vice president, Solar Semiconductor.

Crystalline technology has proven its performance stability over long periods even in field conditions. "One of the scoring factors for C-Si cells is its high efficiency. Two decades ago, crystalline silicon cell efficiency was 8-10 per cent. Today, we are getting 16-17 per cent efficiency," says Sunil Goel, vice president, Maharishi Solar Technology.

Thin-film was seen as the technology of the future when it first appeared in the global markets some years ago since it used a fraction of polysilicon (very expensive at that time, costing as much as \$700/kg) compared to C-Si. However, two things went against thin-film:

1. The price of polysilicon has dropped to one-tenth (\$70/kg), which means the economic argument in favour of thin films is losing its sheen.

2. Thin films have rapid degradation of the power output, especially in the first few years. Over the 25-year lifecycle of the project, C-Si modules degrade at an annual average of 0.5 per cent, whereas a thin-film module degrades at about 1.25 per cent. Due to the long-term nature of the business, this translates

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